

Electro Mechanical Systems (EE2010)

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Course Instructor(s)

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Final Exam

Total Time: 3 Hours

Total Marks: 100

Total Questions: 4

Semester: SP-2024

Campus: Lahore

Dept: Electrical

Engineering

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CLO #3: Analyze Synchronous Generator performance along with special emphasis towards environmental cost of generation

Q1:

[marks 25]

A 2.8 KV, 1.5 MVA, 0.85-PF-lagging, 50-Hz, 16 pole, Y-connected synchronous generator, has a synchronous reactance of 1.2Ω and an armature resistance of 0.2Ω . At 50 Hz, its friction, windage and core losses are 50 KW. The field circuit has a dc voltage of 200 V. The resistance of the field circuit is adjustable over the range from 10Ω to 210Ω .

The generator is providing 300A, at 0.8 PF lagging to a load. The line voltage at load terminals is 2500 V. Answer the following questions.

- Draw a properly labelled equivalent circuit diagram with proper current directions and voltage polarities
- Find Voltage regulation
- Field circuit resistance
- Output torque of prime mover

I_f (A)	2	4	6	8	10
$V_{T,NL}$ (V)	1200	2200	2700	2950	3050

OCC for above generator

CLO #2: Investigate working of a DC Machine

Q2: [marks =25]
 A 25HP, 240V compensated DC shunt motor has armature resistance of 0.4Ω . Its field resistance is 80Ω and adjustable resistance connected in series with field coil can vary from 50Ω to 200Ω . Its no load characteristic curve at a speed of 1800 rpm is tabulated as under.

I_F (A)	0.3	0.6	0.9	1.2	1.4	1.6
E_A (V)	130	230	274	288	292	295

Find out following quantities assuming rated terminal voltage in all cases. .

- No load speed of motor if adjustable resistance connected in series with field coil has been set to a value of 100 ohm.
- The speed at a line current of 65A, with same field current as in part (a)
- Fully labeled Circuit diagram for part (b)
- Output torque in part (b) if mechanical and core losses are 700 watts
- Maximum possible no load speed of the motor

CLO #5: Analyze an Induction motor circuit

Q3: [marks =25]

A 120 HP, 440-V, 50-Hz four-pole Y-connected induction motor is working under steady conditions.

The equivalent circuit parameters are

$$R_1 = 0.075 \Omega \quad R_2 = 0.065 \Omega \quad X_M = 7.2 \Omega$$

$$X_1 = 0.17 \Omega \quad X_2 = 0.17 \Omega$$

$$P_{F\&W} = 1.0 \text{ kW} \quad P_{\text{misc}} = 150 \text{ W} \quad P_{\text{CORE}} = 1.1 \text{ kW}$$

For a slip of 5% and at rated voltage and frequency, find out

- Motor speed
- The stator copper losses
- The air gap power
- Shaft torque

CLO #4: Appraise the existence of a rotating magnetic field and its application to a Synchronous Motor

Q4: [marks =25]

A 415-V, Y-connected, 30 pole, synchronous motor is delivering a power of 30 HP at 0.75 lagging power factor and rated voltage. The field current flowing under these conditions is 4.5 A. Its synchronous reactance is 1.2Ω . Total losses of the motor under these conditions are 2300watts. Assume a linear open-circuit characteristic.

Find the torque angle and internally generated voltage

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How much field current is required to run the motor at a PF of 0.8 leading, keeping everything else constant?

What is the load torque?

Assignment 3

Q2 Assume that the internally generated line voltage in above mentioned generator is adjusted to 2.4 kV. And a delta connected load having per phase Impedance of $18.186 + j10.5$ ohm is connected to the generator. Find out the efficiency and regulation of the system under these conditions. If another load of the same size is connected in parallel to the first load then what should be the new terminal voltage and voltage regulation.

Q3 Repeat question 1 for the rated conditions.

Q4 Consider a 400-V, 12 pole, 60 hz, Y connected synchronous generator having synchronous reactance of 1 ohm and armature resistance of 0.11 ohm. The generator is initially delivering a power of 30 KW at 0.75 PF lagging and rated terminal voltage. find out following

- (a) Initial value of internally generated voltage, torque angle and field current
- (b) The value of field current if the generator has to operate under similar real output power conditions and a PF of 0.75 leading.
- (c) The reactive and apparent power in part (b)
- (d) Input power delivered by the prime mover, if frictional and core losses are 1.1 KW.
- (e) Applied torque, and speed in rpm

The magnetization curve for this machine is shown in Figure below

Magnetization Curve for above machine					
I_f (A)	1	2	3	4	5
E_A (V)	100	230	360	460	550

Problem set one (Transformer)

Q1) A 08-kVA 3000/250-V, 0.8PF lagging, single phase distribution transformer has series impedance referred to the primary of $72 + j245 \Omega$. The components of the excitation branch referred to the secondary side are $R_c = 300 \Omega$ and $X_m = j60 \Omega$. The transformer is delivering rated output at rated power factor and rated secondary voltage to a load. Draw appropriate circuit model and find out

- Voltage regulation
- Applied primary voltage
- Voltage drop across series impedance
- Total losses
- Efficiency
- Input power real, reactive and, apparent
- Impedance seen by supply line
- Draw phasor diagram

Q2) A 50-KVA, 6000/440V transformer has been tested using the open circuit and short circuit tests respectively. The test results have been shown below.

<u>OCT</u>	<u>SCT</u>
5800V	450V
1.8 A	10.5A
340W	800W

Both tests have been performed on HV side. Find and draw circuit model on HV as well as LV side.

Q3) Repeat problem 1 for the transformer of Q2 for a power factor of 0.7 lagging.

Q4) Evaluate Q #1 & 3, using primary referred circuit model with all the quantities, referred to primary side.

Assignment 3

Q1 A 2300-V 1200-kVA 0.85-PF-lagging 60-Hz two pole Y-connected synchronous generator has a synchronous reactance of 1.2Ω and an armature resistance of 0.2Ω . At 50 Hz, its friction and windage losses are 31 kW, and its core losses are 21 kW. The field circuit has a dc voltage of 200 V, and the maximum I_f is 10 A. The resistance of the field circuit is adjustable over the range from 20 to 200 Ω . The generator is providing 280 A to a load at 0.8 power factor and 95% rated voltage. Find out the following quantities.

- 1) Internally generated voltage
- 2) The field current
- 3) The value of field resistance
- 4) Voltage regulation
- 5) Efficiency
- 6) Applied torque by the prime mover
- 7) Rotational speed of prime mover in rpm
- 8) The input power of the prime mover if the efficiency of the prime mover is 30%

(Note: the open circuit characteristics of the generator are shown below)

